

Symmetries and Conservation HW-3

PROBLEM 1. Poisson bracket: In classical mechanics, the Poisson bracket is defined as

$$\{f, g\}_{P.B.} = \sum_i \frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i}$$

where the subscript i labels different generalized coordinates. In the classical field theory, the Poisson bracket can be defined as

$$\{f, g\}_{P.B.} = \sum_i \int d^3\mathbf{x} \left(\frac{\delta f}{\delta \phi_i(\mathbf{x}, t)} \frac{\delta g}{\delta \pi_i(\mathbf{x}, t)} - \frac{\delta f}{\delta \pi_i(\mathbf{x}, t)} \frac{\delta g}{\delta \phi_i(\mathbf{x}, t)} \right)$$

where $\pi_i(\mathbf{x}, t) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_i(\mathbf{x}, t)}$ is the conjugate field (or "momentum") of $\phi_i(\mathbf{x}, t)$, and $\frac{\delta f}{\delta \phi_i(\mathbf{x}, t)}$ is the function derivative of f with respect to the field $\phi_i(\mathbf{x}, t)$. Note that

$$\delta \phi_i(\mathbf{x}, t) / \delta \phi_j(\mathbf{y}, t) = \delta \pi_i(\mathbf{x}, t) / \delta \pi_j(\mathbf{y}, t) = \delta_{ij} \delta^3(\mathbf{x} - \mathbf{y}) \quad \text{and} \quad \delta \phi_i(\mathbf{x}, t) / \delta \pi_j(\mathbf{y}, t) = \delta \pi_i(\mathbf{x}, t) / \delta \phi_j(\mathbf{y}, t) = 0$$

1. (10 points) Calculate $\{\phi_i(\mathbf{x}, t), \pi_j(\mathbf{y}, t)\}$. Thus, the Poisson bracket is very similar to the commutation in quantum mechanics.
2. (15 points) Consider the Lagrangian density of the complex field which respects Schrödinger equation

$$\mathcal{L} = |\partial_\mu \phi|^2 - m^2 |\phi|^2 - \frac{\lambda}{4} |\phi|^4$$

Derive the conserved current j^μ for the continuum $U(1)$ symmetry by Noether theorem.

3. (15 points) In principle, the current j^μ obtained above can be differed by an overall coefficient, i.e. αj^μ is also conserved where α can be an arbitrary constant. To fix this coefficient, we consider the total charge

$$Q = \int d^3\mathbf{x} j^0(\mathbf{x})$$

Prove Q is conserved. Rewrite the expression of j^0 in terms of ϕ (ϕ^*) and the corresponding conjugate fields π_ϕ (π_{ϕ^*}). Calculate the Poisson bracket $\{Q, \phi(\mathbf{x}, t)\}$.

4. (10 points) We require that

$$\{Q, \phi(\mathbf{x}, t)\} = -ie\phi(\mathbf{x}, t)$$

where e is the charge associated with the field ϕ . How to rewrite the current j^μ so that the above Poisson bracket is satisfied?

5. (20 points) Derive the stress-energy tensor $T^{\mu\nu}$ of the Lagrangian density in Eqn. 1 for the translation symmetry.
6. (10 points) Define the 4-dimensional momentum P^μ of the field as

$$P^\mu = \int d^3\mathbf{x} T^{0\mu}$$

Prove it is a conserved quantity.

7. (10 points) Calculate $\{P^\mu, \phi(\mathbf{x}, t)\}_{P.B.}$. Argue why it should be identified as "momentum".
 8. (10 points) Consider N complex scalar fields $\phi_i (i = 1, \dots, N)$, with the Lagrangian density of

$$\mathcal{L} = \sum_{i=1}^N |\partial_\mu \phi_i|^2 - m^2 \sum_{i=1}^N |\phi_i|^2 - \frac{\lambda}{4} \left(\sum_{i=1}^N |\phi_i|^2 \right)^2$$

It is obvious that $\mathcal{L}$ is invariant under $U(N)$. By decomposing each complex field into two real fields, argue that the intrinsic continuous symmetry group should be $SO(2N)$. By counting the numbers of independent matrix elements of $U(N)$ and $SO(2N)$ matrices, which matrix has more degrees of freedom?

SOLUTION:

1.

$$\begin{aligned} \{\phi_i(\mathbf{x}, t), \pi_j(\mathbf{y}, t)\} &= \sum_k \int d^3z \left(\frac{\delta \phi_i(\mathbf{x}, t)}{\delta \phi_k(\mathbf{z}, t)} \frac{\delta \pi_j(\mathbf{y}, t)}{\delta \pi_k(\mathbf{z}, t)} - \frac{\delta \pi_j(\mathbf{y}, t)}{\delta \phi_k(\mathbf{z}, t)} \frac{\delta \phi_i(\mathbf{x}, t)}{\delta \pi_k(\mathbf{z}, t)} \right) \\ &= \sum_k \int d^3z (\delta_{ik} \delta_{jk} \delta^3(\mathbf{x} - \mathbf{z}) \delta^3(\mathbf{y} - \mathbf{z})) = \delta_{ij} \delta(\mathbf{x} - \mathbf{y}) \end{aligned}$$

2.

By Noether's theorem,

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} i\phi - \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} i\phi^* = i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi)$$

3.

Because $\partial_\mu j^\mu = 0$ and j^μ vanishes at infinity,

$$\frac{dQ}{dt} = \int d^3\mathbf{x} \partial_0 j^0(\mathbf{x}) = - \int d^3\mathbf{x} \partial_i j^i(\mathbf{x}) = - \int_{\partial} j^i(\mathbf{x}) \cdot d\mathbf{S} = 0 \quad (i = 1, 2, 3)$$

Thus Q is conserved.

$$j^0 = i(\phi \partial_t \phi^* - \phi^* \partial_t \phi), \pi_\phi = \frac{\partial \mathcal{L}}{\partial(\partial_t \phi)} = \partial_t \phi^*, \pi_{\phi^*} = \partial_t \phi$$

$$\begin{aligned} \{Q, \phi(\mathbf{x}, t)\} &= \int d^3\mathbf{y} \left(\frac{\delta Q}{\delta \phi(\mathbf{y}, t)} \frac{\delta \phi(\mathbf{x}, t)}{\delta \pi_\phi(\mathbf{y}, t)} - \frac{\delta Q}{\delta \pi_\phi(\mathbf{y}, t)} \frac{\delta \phi(\mathbf{x}, t)}{\delta \phi(\mathbf{y}, t)} + \frac{\delta Q}{\delta \phi^*(\mathbf{y}, t)} \frac{\delta \phi(\mathbf{x}, t)}{\delta \pi_{\phi^*}(\mathbf{y}, t)} - \frac{\delta Q}{\delta \pi_{\phi^*}(\mathbf{y}, t)} \frac{\delta \phi(\mathbf{x}, t)}{\delta \phi^*(\mathbf{y}, t)} \right) \\ &= \int d^3\mathbf{y} (-\delta(\mathbf{x} - \mathbf{y}) \frac{\delta Q}{\delta \pi_\phi(\mathbf{y}, t)}) = - \int d^3\mathbf{z} \frac{\delta i(\phi \partial_t \phi^* - \phi^* \partial_t \phi)(\mathbf{z}, t)}{\delta(\partial_t \phi^*(\mathbf{x}, t))} = -i \int d^3\mathbf{z} \phi(\mathbf{z}, t) \delta(\mathbf{x} - \mathbf{z}) = -i\phi(\mathbf{x}, t) \end{aligned}$$

4.

It is apparent that letting $j^\mu \rightarrow e j^\mu$ leads to desired result.

5.

We have known that by symmetry of translation, the stress-energy tensor can be represented as:

$$T^\mu{}_\nu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\nu \phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi^*)} \partial_\nu \phi^* - \delta^\mu{}_\nu \mathcal{L}$$

Plugging in the Lagrangian density gives:

$$T^\mu{}_\nu = \partial^\mu \phi^* \partial_\nu \phi + \partial^\mu \phi \partial_\nu \phi^* - \delta^\mu{}_\nu (|\partial_\mu \phi|^2 - m^2 |\phi|^2 - \frac{\lambda}{4} |\phi|^4)$$

6.

$$P^\mu = \int d^3\mathbf{x} \left(\partial_t \phi^* \partial^\mu \phi + \partial_t \phi \partial^\mu \phi^* - \delta^{0\mu} (|\partial_\mu \phi|^2 - m^2 |\phi|^2 - \frac{\lambda}{4} |\phi|^4) \right)$$

Due to the conservation property $\partial_\mu T^\mu{}_\nu = 0$,

$$\partial_t P^\mu = \int d^3\mathbf{x} \partial_t T^{0\mu} = - \int d^3\mathbf{x} \partial_i T^{i\mu} = - \int_{\partial} T^{i\mu} dS_i = 0$$

Thus P^μ is conserved.

7.

$$P^0 = \int d^3\mathbf{x} (m^2 |\phi|^2 + \frac{\lambda}{4} |\phi|^4 + |\partial_t \phi|^2 - |\partial_i \phi|^2)$$

$$P^i = \int d^3\mathbf{x} (\partial_t \phi^* \partial_i \phi + \partial_t \phi \partial_i \phi^*)$$

$$\{P^0, \phi(\mathbf{x}, t)\} = - \frac{\delta \int d^3\mathbf{z} (m^2 |\phi|^2 + \frac{\lambda}{4} |\phi|^4 + |\partial_t \phi|^2 - |\partial_i \phi|^2)}{\delta(\partial_t \phi^*(\mathbf{x}, t))} = - \int d^3\mathbf{z} \delta(\mathbf{x} - \mathbf{z}) \partial_t \phi(\mathbf{z}, t) = - \partial_t \phi(\mathbf{x}, t)$$

$$\{P^i, \phi(\mathbf{x}, t)\} = - \frac{\delta \int d^3\mathbf{z} (\partial_t \phi^* \partial_i \phi + \partial_t \phi \partial_i \phi^*)}{\delta(\partial_t \phi^*(\mathbf{x}, t))} = \int d^3\mathbf{z} \partial_i \phi(\mathbf{z}, t) \delta(\mathbf{x} - \mathbf{z}) = - \partial_i \phi(\mathbf{x}, t)$$

It is momentum because we can observe

$$[-\partial_\mu, \phi(\mathbf{x}, t)]f = -\partial_\mu(\phi f) + \phi \partial_\mu f = \{P^\mu, \phi(\mathbf{x}, t)\}f$$

So we observe that there is a relationship between P^μ and the generator of translation $-\partial_\mu$, that's why we call P^μ momentum.

8.

Let $\phi_i = \chi_i + i\xi_i$, $\mathbf{V} = (\chi_i, \xi_i)$.

$$\mathcal{L} = \sum_{i=1}^N [(\partial_\mu \chi_i)^2 + (\partial_\mu \xi_i)^2 - m^2(\chi_i^2 + \xi_i^2)] - \frac{\lambda}{4} \left(\sum_{i=1}^N (\chi_i^2 + \xi_i^2) \right)^2 = \|\partial_\mu \mathbf{V}\|^2 - m^2 \|\mathbf{V}\|^2 - \frac{\lambda}{4} \|\mathbf{V}\|^4$$

where $\|\partial_\mu \mathbf{V}\|^2 = \partial_\mu \mathbf{V} \partial^\mu \mathbf{V}$

Consider the $SO(2N)$ transformation:

$$(\chi'_i, \xi'_i)^T = O(\chi_i, \xi_i)^T, O \in SO(2N)$$

then none of the terms in $\mathcal{L}$ changes. Thus $\mathcal{L}$ has $SO(2N)$ symmetry.

DOF:

$U(N): N^2$.

$SO(2N): \frac{2N(2N-1)}{2} = 2N^2 - N$ (because the generator of $SO(2N)$ contains all anti-symmetric matrices)

Apparently $SO(2N)$ has more degrees of freedom.

PROBLEM 2. We have discussed $U(1)$ spontaneous symmetry breaking in the complex scalar field theory with the Lagrangian of

$$\mathcal{L} = |\partial_\mu \phi|^2 + \mu^2 |\phi|^2 - \lambda |\phi|^4$$

where ϕ is a complex scalar field and $\mu^2 > 0$. Assume the ground state is chosen to be $\phi = e^{i\alpha} \phi_0$ with α being an arbitrary real number and $\phi_0 = \sqrt{\mu^2 / (2\lambda)}$. Work out the massive and the Nambu-Goldstone massless modes.

SOLUTION:

Consider the excitation

$$\phi = e^{i\alpha}\phi_0 + \frac{1}{\sqrt{2}}(e^{i\alpha}\chi_1 + ie^{i\alpha}\chi_2)$$

Then the Lagrangian density is:

$$\begin{aligned}\mathcal{L} = & \frac{1}{2}\partial_\mu(\chi_1 + i\chi_2)\partial^\mu(\chi_1 - i\chi_2) + \mu^2(e^{i\alpha}\phi_0 + \frac{\sqrt{2}}{2}(e^{i\alpha}\chi_1 + ie^{i\alpha}\chi_2))(e^{-i\alpha}\phi_0 + \frac{\sqrt{2}}{2}(e^{-i\alpha}\chi_1 - ie^{-i\alpha}\chi_2)) - \\ & \lambda|(e^{i\alpha}\phi_0 + \frac{\sqrt{2}}{2}(e^{i\alpha}\chi_1 + ie^{i\alpha}\chi_2))|^4\end{aligned}$$

Saving all terms that have order less than 2,

$$\begin{aligned}\mathcal{L} = & \frac{1}{2}(|\partial_\mu\chi_1|^2 + |\partial_\mu\chi_2|^2) + \mu^2\phi_0^2 + \frac{\mu^2}{2}\chi_1^2 + \sqrt{2}\mu^2\phi_0\chi_1 + \frac{\mu^2}{2}\chi_2^2 - \lambda\phi_0^2(3\chi_1^2 + \sqrt{2}\phi_0\chi_1 + \chi_2^2) \\ = & (\frac{1}{2}|\partial_\mu\chi_1|^2 - \mu^2\chi_1^2) + \frac{1}{2}|\partial_\mu\chi_2|^2 + \mu^2\phi_0^2\end{aligned}$$

And we see that χ_2 is massless, and χ_1 has mass $m_{\chi_1} = \sqrt{2}\mu$.

PROBLEM 3. Counting the number of Goldstone modes: In class, we have discussed the emergence of Goldstone modes due to spontaneously breaking of continuous symmetries by the ground state. In this problem, we study how to relate the number of Goldstone modes with the generators of Lie group/algebra by explicitly working out an example on partial symmetry breaking. Suppose we have three real scalar fields $\varphi^a (a = 1, \dots, 3)$ with the Lagrangian of

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\varphi^a)(\partial^\mu\varphi^a) - V(\varphi) \quad \text{with} \quad V(\varphi) = -\frac{\mu^2}{2}\varphi^a\varphi^a + \frac{\lambda}{4}(\varphi^a\varphi^a)^2$$

where $\mu^2 > 0$.

1. (10 points) Identify the symmetry group of $\mathcal{L}$, and find out all possible ground states.
2. (10 points) Suppose the ground state $\vec{\varphi}_0$ is chosen so that $\varphi^1 = \varphi^2 = 0$, identify the broken and unbroken symmetries, and their generators. Also expand the Lagrangian near $\vec{\varphi}_0$.
3. (10 points) Find out the mass of the massive mode(s). Identify the Nambu-Goldstone massless mode(s). How many massless modes?
4. (10 points) Can you find the relation between the number of massless modes and the broken symmetries?

SOLUTION:

1.

Quite apparently the symmetry is $U \in O(3)$ such that $\mathcal{L}[\varphi, \partial_\mu\varphi] = \mathcal{L}[U(\varphi), U(\partial_\mu\varphi)]$

ground states are apparently $\partial_\mu\phi^\alpha = 0, ||\varphi||^2 = \sqrt{\frac{\mu^2}{\lambda}}$

2.

Broken symmetry: $O(3)/O(1)$ (here means quotient set (or quotient space $\mathfrak{o}(3)/\mathfrak{o}(1)$), not quotient group)

unbroken symmetry: $O(1)$ (rotation with respect to $\vec{\varphi}_0$)

Generator of $O(3)$: $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Generator of $O(1)$: $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Suppose $\varphi_\alpha = \vec{\varphi}_0 + \chi_\alpha$, then save all terms in $\mathcal{L}$ to order 2:

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}(\partial_\mu \chi_\alpha)^2 + \frac{\mu^2}{2}(|\chi|^2 + \varphi_0^2 + 2\varphi_0 \chi_1) - \frac{\lambda}{4}(2\varphi_0^2 |\chi|^2 + \varphi_0^4 + 4\varphi_0^2 \chi_1^2 + 4\varphi_0^3 \chi_1) \\ &= \frac{\mu^4}{2\lambda} + \frac{1}{2}(\partial_\mu \chi_\alpha)^2 + \mu^2 \varphi_0 \chi_1 - \mu^2 \chi_1^2 - \mu^2 \varphi_0 \chi_1 = \frac{\mu^4}{2\lambda} + \frac{1}{2}(\partial_\mu \varphi_\alpha)^2 - \mu^2 \chi_1^2 \end{aligned}$$

3.

massive mode: $\chi_1, m_{\chi_1} = \sqrt{2}\mu$.

massless modes: χ_2, χ_3 . There are 2 massless modes.

4.

numbers of massless modes = dimension of broken symmetries

PROBLEM 4. Consider the Lagrangian of sourceless EM field

$$\mathcal{L}_{EM} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1. (10 points) Write down the canonical energy-momentum tensor $T_c^{\mu\nu}$ obtained from the translation symmetry. You may find the tensor is not gauge-invariant, implying this tensor is unphysical.
2. (10 points) Modify this tensor $T_c^{\mu\nu} \rightarrow T^{\mu\nu}$ so that the difference is a total derivative ($T_c^{\mu\nu} - T^{\mu\nu} = \partial_\lambda f^{\mu\nu\lambda}$ suggesting the integral over the whole space-time is invariant). In addition, the new tensor $T^{\mu\nu}$ should be conserved ($\partial_\mu T^{\mu\nu} = 0$), symmetric ($T^{\mu\nu} = T^{\nu\mu}$), and gauge-invariant.
3. (10 points) Express T^{00} and T^{0i} in terms of $\mathbf{E}$ and $\mathbf{B}$ fields. You should find it is the energy/momentum density of the EM field.
4. (10 points) Obtain $T_c^{\mu\nu}$ for the scalar electrodynamics with the Lagrangian of

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + |D_\mu \psi|^2 - m^2 |\psi|^2$$

where $D_\mu = \partial_\mu - ieA_\mu$. Is it gauge invariant? If not, modify this energy-momentum tensor so that it is also conserved, symmetric, and gauge-invariant. Explicitly show that the difference between two tensors is a total derivative, i.e. $T_c^{\mu\nu} - T^{\mu\nu} = \partial_\lambda h^{\mu\nu\lambda}$. (In this case, note $\partial_\mu F^{\mu\nu} \neq 0$.)

SOLUTION:

1.

$$T_c^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\rho)} \partial^\nu A_\rho - \eta^{\mu\nu} \mathcal{L} = \frac{1}{4} \eta^{\mu\nu} F_{\rho\theta} F^{\rho\theta} - F^{\mu\rho} \partial^\nu A_\rho$$

$T_c^{\mu\nu}$ is not gauge invariant because when $A^\mu \rightarrow A^\mu + \partial^\mu G$,

$$T_c^{\mu\nu} \rightarrow T_c^{\mu\nu} - F^{\mu\rho} \partial^\nu \partial_\rho G$$

2.

Notice that $F^{\mu\nu}\partial^\nu\partial_\rho G = F^{\mu\nu}\partial_\rho\partial^\nu G$, then let

$$T^{\mu\nu} = T_c^{\mu\nu} + F^{\mu\rho}\partial_\rho A^\nu = \frac{1}{4}\eta^{\mu\nu}F_{\rho\theta}F^{\rho\theta} - F^{\mu\rho}(\partial^\nu A_\rho - \partial_\rho A^\nu) = \frac{1}{4}\eta^{\mu\nu}F_{\rho\theta}F^{\rho\theta} - F^{\mu\rho}F^\nu{}_\rho$$

It is conserved because $\partial_\mu T^{\mu\nu} = \partial_\mu T_c^{\mu\nu} + \partial_\mu F^{\mu\rho}\partial_\rho A^\nu = 0$ by equation of motion $\partial_\mu F^{\mu\rho} = 0$.

It is apparently symmetric.

It is gauge invariant because $F^{\mu\nu}$ is gauge invariant.

3.

$$T^{00} = \frac{1}{4}F_{\rho\theta}F^{\rho\theta} - F^{0\rho}F^0{}_\rho = \frac{1}{2}(-E^2 + B^2) + E^\rho E_\rho = \frac{1}{2}(E^2 + B^2)$$

$$T^{0i} = -F^{0\rho}F^i{}_\rho = E^\rho\epsilon_{i\rho j}B^j = \epsilon_{\rho ji}E^\rho B^j = [\mathbf{E} \times \mathbf{B}]_i$$

4. Let $T_{field}^{\mu\nu}$ be the energy momentum tensor in 2, which is gauge invariant.

$$\partial_\mu T_{field}^{\mu\nu} = -j^\rho F_\rho{}^\nu$$

consider $T_{matter}^{\mu\nu}$:

$$T_{matter}^{\mu\nu} = (D^\mu\psi)^*\partial^\nu\psi + D^\mu\psi\partial^\nu\psi^* - \eta^{\mu\nu}(|D_\mu\psi|^2 - m^2|\psi|^2)$$

Then

$$\begin{aligned}\partial_\mu T_{matter}^{\mu\nu} &= (\partial_\mu D^\mu\psi^*)\partial^\nu\psi + (D^\mu\psi)^*\partial_\mu\partial^\nu\psi + c.c. - [(\partial^\nu D_\mu\psi)(D^\mu\psi)^* + D_\mu D^\mu\psi^*(\partial^\nu\psi) + c.c.] \\ &= ie(D^\mu\psi)^*\partial^\nu(A_\mu\psi) - ieA_\mu(D^\mu\psi)^*(\partial^\nu\psi) + c.c. = ie\psi(D^\mu\psi)^*\partial^\nu A_\mu + c.c. = (\partial^\nu A_\mu)j^\mu\end{aligned}$$

Thus, let $T_c^{\mu\nu} = T_{matter}^{\mu\nu} + T_{field}^{\mu\nu} - A^\nu j^\mu$, then $\partial_\mu T_c^{\mu\nu} = -j^\mu F^\nu{}_\mu + (\partial^\nu A_\mu)j^\mu - \partial_\mu(A^\nu j^\mu) = 0$ because $\partial_\mu j^\mu = 0$.

We claim that $T_c^{\mu\nu}$ already is gauge invariant. Consider gauge transform $A_\mu \rightarrow A_\mu + \frac{\partial_\mu\theta}{e}$, $\psi \rightarrow e^{i\theta}\psi$, then

$$T_{matter}^{\mu\nu} \rightarrow T_{matter}^{\mu\nu} + i\partial^\nu\theta\psi(D^\mu\psi)^* - i\partial^\nu\theta\psi^*D^\mu\psi = T_{matter}^{\mu\nu} + \frac{\partial^\nu\theta}{e}j^\mu$$

$$T_{field}^{\mu\nu} \rightarrow T_{field}^{\mu\nu}$$

$$-A^\nu j^\mu \rightarrow -A^\nu j^\mu - \frac{\partial^\nu\theta}{e}j^\mu$$

And we see that $T_c^{\mu\nu}$ is gauge invariant. □